Zhikai Chen

Contact

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Research Interests

- LLM agents, with focus on memory, context engineering, post-training, and agentic search
- Forecasting and predictive foundation models
- Relational deep learning and foundation models for relational data

Experience

Amazon AWS, Applied Scientist Intern 05/2026 – 08/2026
- Topic: Red teaming post-training for agents

Amazon AWS, Applied Scientist Intern 05/2025 – 12/2025
- Outcome: Relatron (ICLR 2026) and RFMBench

Google, Student Researcher 03/2025 – 05/2025
- Outcome: Relational foundation model research

Amazon AWS, Applied Scientist Intern 06/2024 – 09/2024
- Outcome: AutoG (ICLR 2025)

Baidu, Research Intern 05/2023 – 09/2023
- Outcome: "Exploring the Potential of LLMs" (SIGKDD Explorations, 550+ citations)

Selected Works

1. "Exploring Cross-Scenario Generality of Agentic Memory Systems: Diagnostics and a Strong Baseline"
Zhikai Chen, Jialiang Gu, Junyu Yin, Xianxuan Long, Shenglai Zeng, Xiaoze Liu, Kai Guo, Keren Zhou, Jiliang Tang. **IJCNLP-AACL (main)**, 2026.
Keywords: LLM Agents, Memory Systems

2. "OpenRFM: Dissecting Relational In-Context Learning"
Zhikai Chen, Junyu Yin, Jialiang Gu, Siheng Xiong, Xiaoze Liu, Ruowang Zhang, Keren Zhou, Kai Guo. Preprint, 2026.
Keywords: Relational Foundation Models, In-Context Learning

3. "Relatron: Automating Relational Machine Learning over Relational Databases"
Zhikai Chen, Han Xie, Jian Zhang, Jiliang Tang, Xiang Song, Huzefa Rangwala. **ICLR**, 2026.
Keywords: Relational Deep Learning, AutoML, Theory

4. "AutoG: Towards Automatic Graph Construction from Tabular Data"
Zhikai Chen, Han Xie, Jian Zhang, Xiang Song, Jiliang Tang, Huzefa Rangwala, George Karypis. **ICLR**, 2025.
Keywords: DSAgent, Relational Database

5. "RFMBench: Towards Principled Benchmarking of Relational Foundation Models"
Zhikai Chen*, Han Xie, Jian Zhang, Haiyang Yu, Huzefa Rangwala, Xiang Song. Under review.
Keywords: Relational Foundation Models

6. "Text-space Graph Foundation Models: Comprehensive Benchmarks and New Insights"
Zhikai Chen, Haitao Mao, Jingzhe Liu, Yu Song, Bingheng Li, Wei Jin, Bahare Fatemi, Anton Tsitsulin, Bryan Perozzi, Hui Liu, Jiliang Tang. **NeurIPS**, 2024.
Keywords: Graph Foundation Models, Benchmarks

7. "Label-free Node Classification on Graphs with Large Language Models (LLMs)"
Zhikai Chen, Haitao Mao, Hongzhi Wen, Haoyu Han, Wei Jin, Haiyang Zhang, Hui Liu, Jiliang Tang. **ICLR**, 2024.
Keywords: LLM for Crowdsourcing, Data Annotation

8. "Exploring the Potential of Large Language Models (LLMs) in Learning on Graphs"
Zhikai Chen, Haitao Mao, Hang Li, Wei Jin, Hongzhi Wen, Xiaochi Wei, Shuaiqiang Wang, Dawei Yin, Wenqi Fan, Hui Liu, Jiliang Tang. **SIGKDD Explorations**, 2023. (550+ citations)
Keywords: Large Language Models, Graph Machine Learning

9. "Graph Foundation Models"
   Haitao Mao*, **Zhikai Chen***, Wenzhuo Tang, Jianan Zhao, Yao Ma, Tong Zhao, Neil Shah, Michael Galkin, Jiliang Tang (* Equal Contributions). **ICML Spotlight**, 2024.
   Keywords: Graph Foundation Models, Pretraining

Funding

- NSF ACCESS, 1,500,000 computation units 2026
- Tinker Research Grant, $5,000 2026

Projects

1. **Vibe Research** [Code] — An agentic system designed for human-in-the-loop automatic research.
2. **AutoG** [Code] — LLM-based data agent for automatic graph schema generation from data lakes. Internal adoption and demonstration at AWS re:Invent.
3. **Graph-LLM** [Code] — Exploring how LLMs can help learning on graphs: LLM-as-enhancer, LLM-as-predictor, LLM-as-annotator. 300+ stars.
4. **LLM-GNN** [Code] — LLMs as annotators for node classification, superior performance without human supervision. 80+ stars.
5. **ML PhD Interview Notes** [Code] — A Hitchhiker's Guide to ML PhD Job Hunting, covering ML/LLM interview preparation across ML systems, LLM agents, coding, system design, and algorithms. 40+ stars.

Education

Michigan State University, East Lansing, USA 01/2023 – Present
- Ph.D., Computer Science and Engineering. Advisor: Foundation Professor Jiliang Tang

Shanghai Jiao Tong University, Shanghai 09/2020 – 01/2023
- M.Eng., Cybersecurity. Advisor: Professor Li Pan

Shanghai Jiao Tong University, Shanghai 09/2016 – 08/2020
- B.Eng., ECE, Joint Institute. Advisor: Professor Xudong Wang, IEEE Fellow

Publication

Conference and Journal Papers

17. "Relatron: Automating Relational Machine Learning over Relational Databases"
    Zhikai Chen, Han Xie, Jian Zhang, Jiliang Tang, Xiang Song, Huzefa Rangwala
    ICLR, 2026
16. "Exploring Cross-Scenario Generality of Agentic Memory Systems: Diagnostics and a Strong Baseline"
    Zhikai Chen, Jialiang Gu, Junyu Yin, Xianxuan Long, Shenglai Zeng, Xiaoze Liu, Kai Guo, Keren Zhou, Jiliang Tang
    IJCNLP-AACL (main), 2026
15. "A Pre-training Framework for Relational Data with Information-theoretic Principles"
    Quang Truong, **Zhikai Chen**, Mingxuan Ju, Tong Zhao, Neil Shah, Jiliang Tang
    NeurIPS, 2025
14. "Unveiling Mode Connectivity in Graph Neural Networks"
    Bingheng Li, **Zhikai Chen**, Haoyu Han, Shenglai Zeng, Jingzhe Liu, Jiliang Tang
    KDD, 2025
13. "Graph Machine Learning in the Era of Large Language Models (LLMs)"
    Wenqi Fan, Shijie Wang, Jiani Huang, **Zhikai Chen**, Yu Song, Wenzhuo Tang, Haitao Mao, Hui Liu, Xiaorui Liu, Dawei Yin, Qing Li
    ACM TIST, 2025
12. "Enhancing ID and Text Fusion via Alternative Training in Session-based Recommendation"
    Juanhui Li, Haoyu Han, **Zhikai Chen**, Harry Shomer, Wei Jin, Amin Javari, Jiliang Tang
    AACL (poster), 2025
11. "AutoG: Towards automatic graph construction from tabular data"
    Zhikai Chen, Han Xie, Jian Zhang, Xiang Song, Jiliang Tang, Huzefa Rangwala, George Karypis
    ICLR (poster), 2025

10. "Improving Causal Reasoning in Large Language Models: A Survey"
Longxuan Yu, Delin Chen, Siheng Xiong, Qingyang Wu, Dawei Li, **Zhikai Chen**, Xiaoze Liu, Liangming Pan
NAACL (main), 2025

9. "A Pure Transformer Pretraining Framework on Text-attributed Graphs"
Yu Song, Haitao Mao, Jiachen Xiao, Jingzhe Liu, **Zhikai Chen**, Wei Jin, Carl Yang, Jiliang Tang, Hui Liu
LOG (poster), 2024.

8. "Neural Scaling Laws on Graphs"
Jingzhe Liu, Haitao Mao, **Zhikai Chen**, Tong Zhao, Neil Shah, Jiliang Tang
LOG (poster), 2024.

7. "Text-space Graph Foundation Model: Comprehensive Benchmarks and New Insights"
Zhikai Chen, Haitao Mao, Jingzhe Liu, Yu Song, Bingheng Li, Wei Jin, Bahare Fatemi, Anton Tsitsulin, Bryan Perozzi, Hui Liu, Jiliang Tang
NeurIPS (poster), 2024.

6. "Graph Foundation Models"
Haitao Mao*, **Zhikai Chen***, Wenzhuo Tang, Jianan Zhao, Yao Ma, Tong Zhao, Neil Shah, Michael Galkin, Jiliang Tang (* Equal Contributions)
ICML (spotlight), 2024

5. "Label-free Node Classification on Graphs with Large Language Models (LLMS)"
Zhikai Chen, Haitao Mao, Hongzhi Wen, Haoyu Han, Wei Jin, Haiyang Zhang, Hui Liu, Jiliang Tang
ICLR (poster), 2024

4. "Exploring the potential of large language models (llms) in learning on graphs"
550+ citations
Zhikai Chen, Haitao Mao, Hang Li, Wei Jin, Hongzhi Wen, Xiaochi Wei, Shuaiqiang Wang, Dawei Yin, Wenqi Fan, Hui Liu, Jiliang Tang
SIGKDD Explorations, 2023.

3. "The 1st International Workshop on Graph Foundation Models (GFM)"
Hosted workshop
Haitao Mao, Jianan Zhao, Xiaoxin He, **Zhikai Chen**, Qian Huang, Zhaocheng Zhu, Jian Tang, Micheal Bronstein, Xavier Bresson, Bryan Hooi, Haiyang Zhang, Xianfeng Tang, Luo Chen, Jiliang Tang
Companion Proceedings of the ACM on Web Conference, 2024.

2. "Demystifying Structural Disparity in Graph Neural Networks: Can One Size Fit All?"
Haitao Mao, **Zhikai Chen**, Wei Jin, Haoyu Han, Yao Ma, Tong Zhao, Neil Shah, Jiliang Tang
NeurIPS (poster), 2023.

1. "Fake News Detection with Contrastive Heterogeneous Graph Representation Learning"
Zhikai Chen, Peng Wu, Li Pan
IJCNN (poster), 2022.

Preprints and Submissions

6. "MemTrace: Probing What Final Accuracy Misses in Long-Term Memory"
Xianxuan Long, **Zhikai Chen**, Shenglai Zeng, Shouren Wang, Kai Guo, Jiliang Tang

5. "OpenRFM: Dissecting Relational In-Context Learning"
Zhikai Chen, Junyu Yin, Jialiang Gu, Siheng Xiong, Xiaoze Liu, Ruowang Zhang, Keren Zhou, Kai Guo

4. "Do Proactive Agents Really Need an LLM to Decide When to Wake and What to Anchor?"
Xiaoze Liu, Ruowang Zhang, Amir H. Abdi, Michel Galley, **Zhikai Chen**, Siheng Xiong, Xiaoqian Wang, Jing Gao

3. "RFMBench: Towards Principled Benchmarking of Relational Foundation Models"
Zhikai Chen*, Han Xie, Jian Zhang, Haiyang Yu, Huzefa Rangwala, Xiang Song

2. "One Model for One Graph: A New Perspective for Pretraining with Cross-domain Graphs"
   Jingzhe Liu, Haitao Mao, **Zhikai Chen**, Bingheng Li, Wenqi Fan, Mingxuan Ju, Tong Zhao, Neil Shah, Jiliang Tang
1. "Learning on Graphs with Large Language Models (LLMs): A Deep Dive into Model Robustness"
   Kai Guo, Zewen Liu, **Zhikai Chen**, Hongzhi Wen, Wei Jin, Jiliang Tang, Yi Chang

Proposal Writing

- Large-scale Graph Foundation Model
  PI: Dr. Jiliang Tang, Dr. Wei Jin
  Role: Drafted part of the proposal
  Result: Computation credits funded by NVIDIA.
- Graph Foundation Models
  PI: Dr. Jiliang Tang
  Role: Designed and drafted part of the proposal
  Result: Funded by Amazon, Meta.

Presentations

Invited Talks

- "Status of graph foundation models" 05/2025
  Presentation at National University of Singapore
- "From LLM4Graph to principled graph foundation models" 02/2024
  Presentation at LOGS Seminar
- "Applying LLMs to solve graph-related problems" 10/2023
  Presentation at AITIME
- "Exploring the potential of LLMs on graphs" 08/2023
  Presentation at Shandong University

Services

Conference Reviewer

- ICML 2025, 2026
- NeurIPS 2024, 2025
- ICLR 2025, 2026
- SIGKDD 2023–2026

Journal Reviewer

- TKDE, TKDD, JMLR, TNNLS